



Sri Lanka Institute of Information Technology

Assignment 2

Report

Data Warehousing and Business Intelligence [IT3021]

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Data Source

The data source used for Assignment 2 is the data warehouse developed in Assignment 1, based on the Hotel Booking Dataset.

The dataset contains 119,390 booking records across 30 hotels located in 15 cities. It includes customer details, booking information, hotel attributes, and transaction-related data.

The data warehouse follows a star schema, with fact_bookings as the central fact table connected to multiple dimensions such as dim_hotel, dim_customer, and dim_date.

This structure enables efficient analytical processing and supports OLAP operations.

Original data set link: [hotel-booking-reservation](https://www.kaggle.com/datasets/genekay/hotel-booking-reservation)

ER Diagram

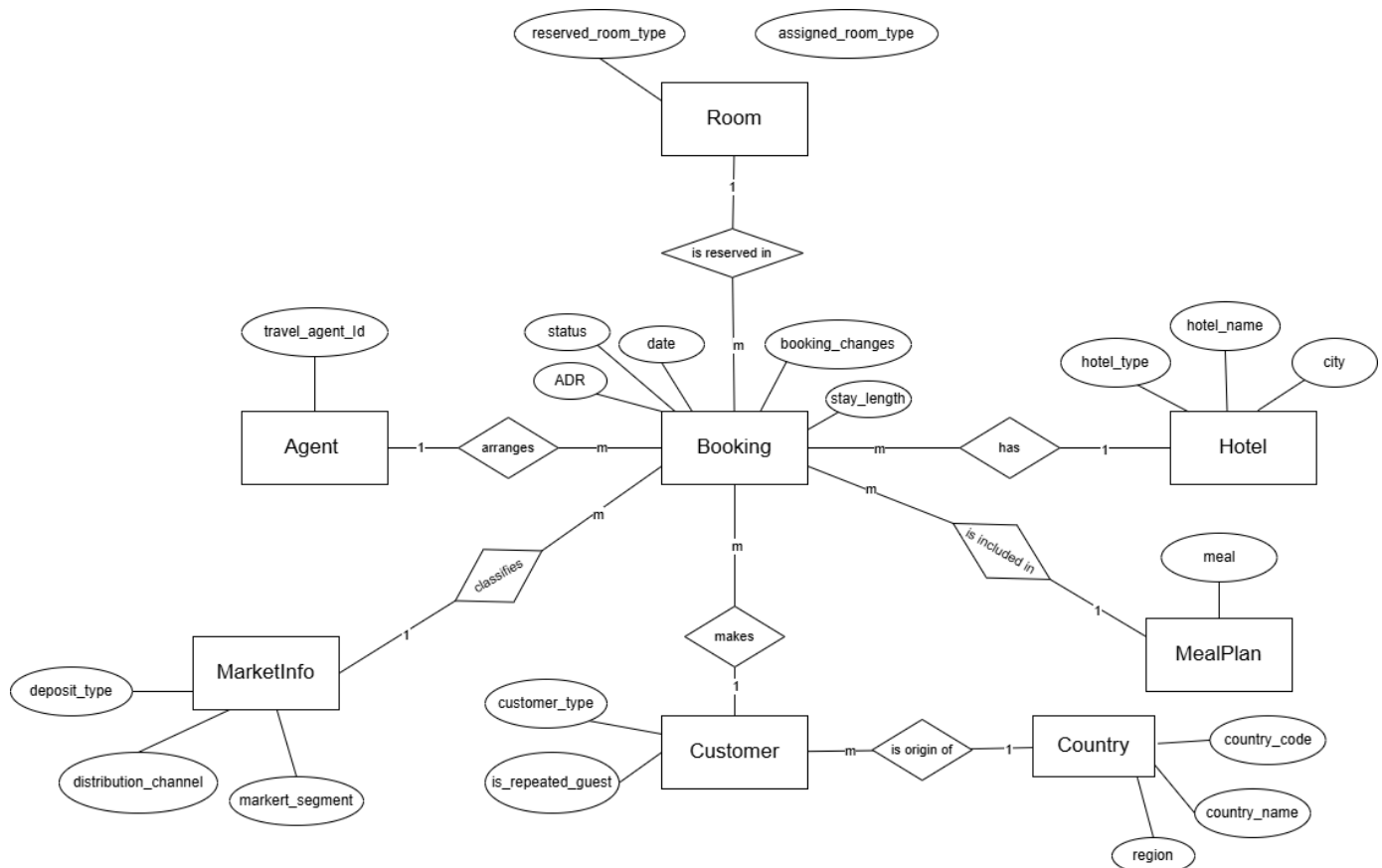


Figure 1 - ER Diagram

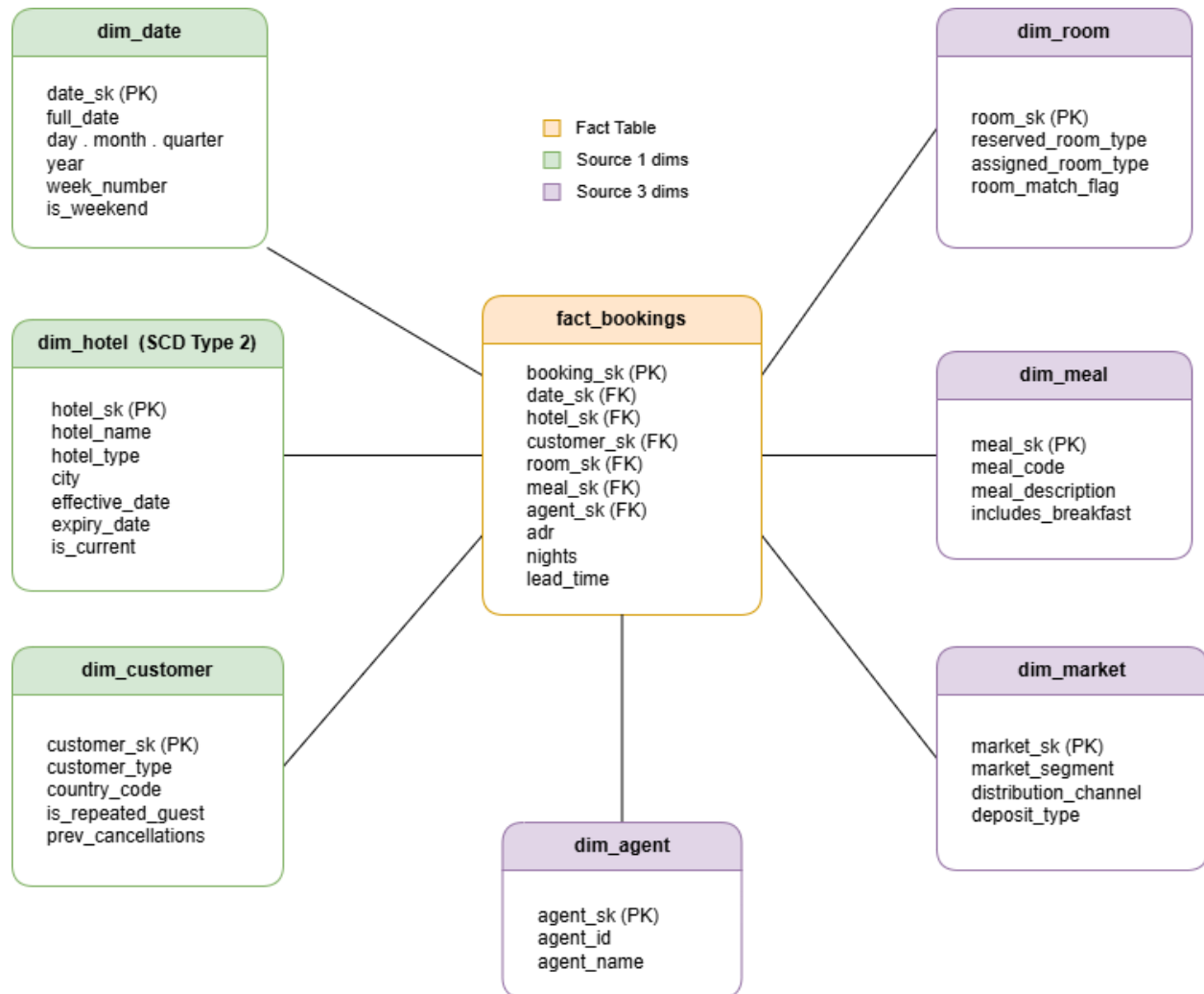
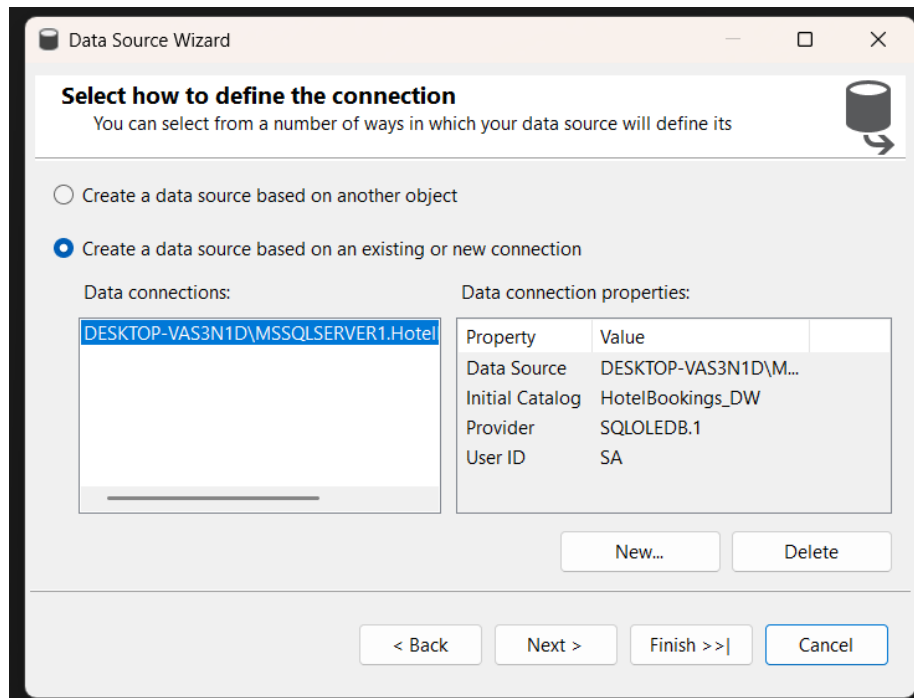


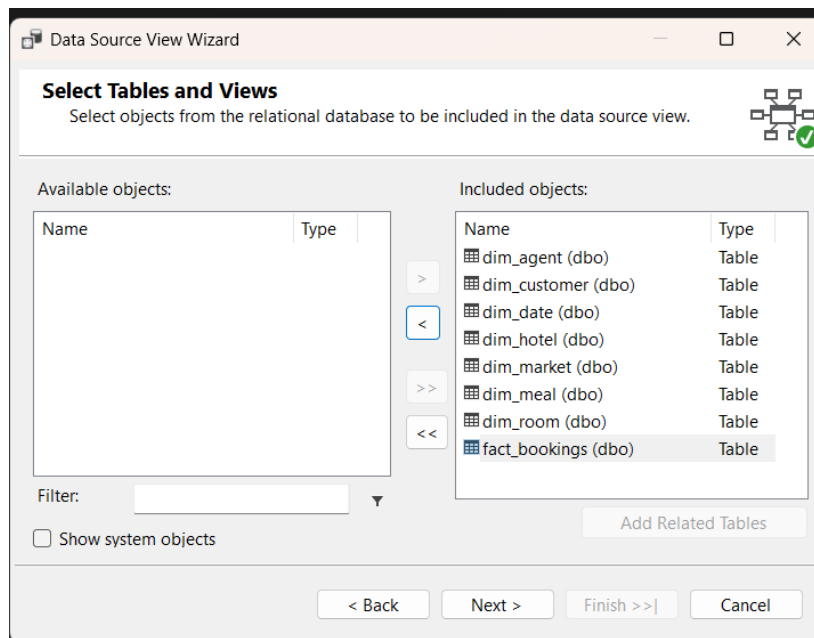
Figure 2 - Star Schema for Hotel Booking DW

SSAS Cube Implementation

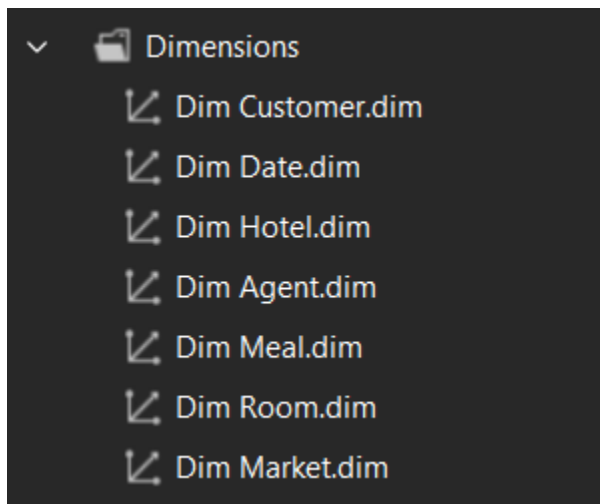
Creating Data Source



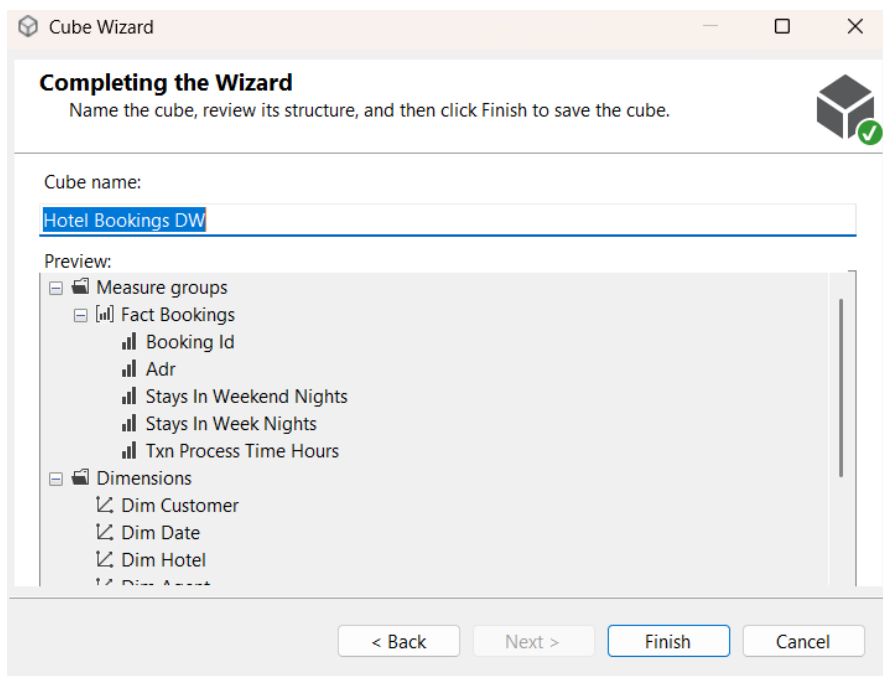
Creating Data Source View



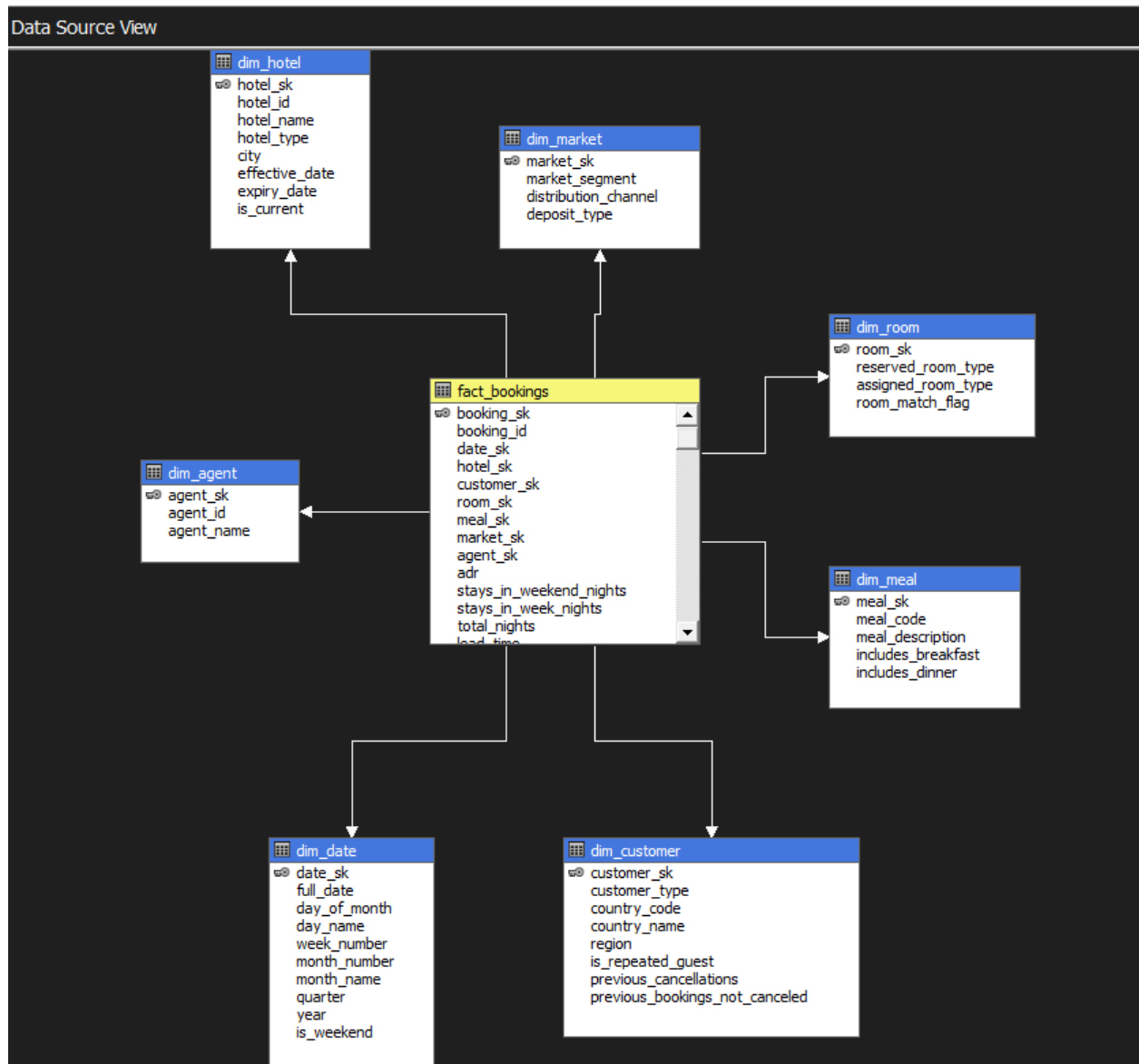
Creating dimensions



Creating cube



Cube Structure



Creating Data Cubes

Creating KPI

Creating a Role

Browsig Cube data

Demonstration of OLAP Operations

Roll-up

Drill-down

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